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Re: Response to Request for Information on the Development of an AI Action Plan

To Whom It May Concern:

I appreciate the opportunity to contribute to the development of a national AI Action Plan that fosters the innovation necessary to sustain and strengthen America's leadership in artificial intelligence. As the founder of Rikka, a firm specializing in AI data protection law and technology transactions, I regularly publish thought leadership on this topic in publications like Forbes.com and Bloomberg Law, as well as counsel businesses on AI development and deployment across financial services, healthcare, and critical infrastructure sectors. I utilize a practical, business-friendly approach that allows private sector companies to get their products to market while complying with applicable laws and regulations.

I recommend that America's AI Action Plan cultivate an ecosystem that fuels innovation while upholding strong governance in privacy, cybersecurity, and intellectual property protection. Just as leading technology companies balance risk management with agility, the U.S. government should adopt targeted, flexible regulations that encourage AI advancements without stifling entrepreneurship. By implementing clear, adaptable policies, incentivizing responsible AI development, and streamlining regulatory frameworks, the U.S. can maintain its position as the global leader in AI.

America's AI leadership requires a balanced regulatory approach—one that avoids the rigid constraints of the European Union AI Act, the prescriptive nature of the Biden Administration's Executive Order, and the state-controlled AI model of China. The U.S. must pursue a middle path that fosters innovation, economic competitiveness, and responsible governance, ensuring AI development aligns with democratic values. Without a pragmatic and adaptable regulatory framework, the U.S. risks losing its competitive edge and ceding AI leadership to nations with less transparent or more restrictive policies.

This RFI response first outlines why privacy, cybersecurity, and intellectual property protections are essential to a robust national AI policy, detailing key risks and challenges. We then present targeted policy recommendations that balance innovation and regulation. Finally, we emphasize the need for a flexible, risk-based regulatory framework that avoids overly restrictive or centralized AI governance models while fostering a thriving AI ecosystem.

I. Overview of Privacy, Cybersecurity, and Intellectual Property Risks in AI

A. Cybersecurity Threats in AI Systems

AI systems can introduce significant cybersecurity risks due to their reliance on vast datasets, complex model architectures (with sometimes limited explainability), and extensive integrations with external systems. AI models are vulnerable to a variety of security risks, including:

- <u>Adversarial Attacks</u>: AI systems can be manipulated using adversarial inputs designed to exploit weaknesses in machine learning models.

- <u>Data Poisoning</u>: Attackers can inject manipulated data into AI training sets, causing models to learn incorrect behaviors or biases that can be exploited post-deployment.

- <u>Model Extraction and Theft</u>: AI models often represent significant intellectual property investments, yet they are susceptible to reverse engineering, allowing adversaries to steal proprietary AI capabilities and data.

- <u>Prompt Injection Attacks</u>: Malicious actors can craft deceptive prompts that cause unintended or incorrect outputs.

- Supply Chain Vulnerabilities: Many AI systems rely on third-party datasets, cloud platforms, and open-source models, creating potential security gaps.

B. Why This Matters

AI introduces sophisticated cybersecurity challenges that, if left unaddressed, erode public trust and provide avenues for foreign and domestic adversaries to manipulate American businesses, policy, and society.

Foreign actors, particularly China and Russia, have exploited AI-powered disinformation campaigns to sow division within American society and undermine democratic institutions. Reports indicate that foreign nations have used AI-generated deepfake technology to spread divisive content and manipulate political discourse. Additionally, nation state linked cyber groups have been implicated in cyberattacks on U.S. critical infrastructure, targeting military supply chains and national security assets to disrupt U.S. responses to geopolitical conflicts. Without robust AI cybersecurity safeguards, these vulnerabilities could be weaponized to influence elections, destabilize markets, and erode institutional trust.

This vulnerability mirrors challenges faced in the cryptocurrency industry, where weak cybersecurity and high-profile breaches—including hacks, exchange collapses, and stolen digital assets—have fueled skepticism, slowed adoption, and invited regulatory scrutiny. Similarly, if AI security risks are not proactively addressed, cyberattacks, model theft, and adversarial AI manipulations could diminish public confidence in AI applications and stifle broader industry adoption. Just as strong security frameworks and fraud prevention mechanisms have been essential in stabilizing the cryptocurrency industry, robust AI security protocols are necessary to protect critical AI infrastructure, maintain public trust, and ensure AI remains a transformative and reliable technology.

II. Privacy Risks in AI Systems

A. Privacy and Surveillance Risks in AI Systems

AI models process vast amounts of data, including potentially sensitive personal data, raising critical concerns about data privacy, surveillance, and user consent. Key risks include:

- Re-identification Risks: AI models trained on anonymized datasets can often re-identify individuals by correlating disparate pieces of information, undermining traditional data protection techniques.

- Inferred Data Exposure: AI systems can generate highly sensitive insights about individuals even without direct access to personal data, raising concerns about implicit privacy violations.

- Data Retention and Compliance Challenges: The long-term storage of training data and model outputs raises concerns under privacy laws like the GDPR, CCPA, and emerging U.S. state AI laws, which require strict data minimization and purpose limitation.

- Profiling and Automated Decision-Making: AI-driven profiling, used in hiring, lending, and law enforcement, often lacks transparency and can lead to discriminatory and biased outcomes. When taken to an extreme, AI-driven profiling can become a precursor to mass surveillance systems that undermine democratic freedoms, mirroring aspects of China's social credit system.

B. Why This Matters

Without robust privacy safeguards, American consumers will vote with their feet, gravitating away from AI models that expose, misuse, or inadequately protect their data. Public trust is a foundational pillar for the success of AI, and models that fail to uphold strong privacy protections will struggle to achieve widespread adoption.

Moreover, the effectiveness of AI models relies heavily on access to diverse, high-quality datasets for training and fine-tuning. Without a significant user base willing to engage with AI platforms due to privacy concerns, American AI developers will face data scarcity, leading to models that are less accurate, less reliable, and ultimately less competitive on the global stage.

A key factor affecting AI model performance is the language and cultural composition of training data. Many leading large language models (LLMs) are trained primarily on English-based datasets derived from a Western cultural perspective, which contributes to their stronger performance in English but reduced effectiveness in other languages. Research highlights that LLMs often struggle with grammar, syntax, and idiomatic expressions in non-English languages, particularly those with different linguistic structures, lower resource availability, or more complex morphology. This linguistic imbalance limits the ability of American AI models to effectively scale and compete in global markets, particularly where local language fluency and cultural nuances are critical for adoption.

Furthermore, languages with fewer digital resources—such as many African, Middle Eastern, and Southeast Asian languages—often suffer from data scarcity, making it difficult to develop AI systems that can serve these populations effectively. If the U.S. does not address this gap, AI models developed in multilingual regions could gain an advantage in global AI adoption by better serving non-English markets.

To maintain American dominance in AI, it is crucial to ensure continued access to diverse, multilingual training data sourced from a broad and representative user base. Restrictive data collection policies or fragmented privacy laws that limit access to training data in both English and other languages risk hampering AI performance and international competitiveness.

If users do not feel confident that their data is being handled responsibly, they may shift toward alternative platforms that prioritize privacy, including AI models developed in jurisdictions with stricter data protection frameworks. This shift would not only weaken U.S. AI leadership but could also drive innovation and economic benefits away from the domestic market.

III. Intellectual Property Risks and the Chilling Effect on AI Innovation

A. Weak Intellectual Property Protection in AI Systems

The United States has long been a leader in global innovation, in part because its Constitution enshrines the principle that inventors and creators should reap the rewards of their innovations. Strong intellectual property (IP) protections serve as a cornerstone of economic growth, incentivizing investment in cutting-edge technologies and promoting fair competition.

The AI sector is particularly vulnerable to unauthorized replication, model theft, and reverse engineering. A notable example is China's DeepSeek, which allegedly used model distillation techniques to reverse engineer OpenAI's ChatGPT. If left unchecked, such activities may allow foreign competitors to replicate and profit from American AI innovations without bearing the costs and risks of original development. This weakens U.S. economic competitiveness, discourages investment in new AI models, and enables adversarial nations to accelerate their own AI capabilities using stolen or distilled intellectual property.

B. Why This Matters

Beyond direct theft, current uncertainties in patent eligibility, copyright ownership, and trade secret protection pose additional risks that could stifle AI-driven progress. Key concerns include:

- Patent Uncertainty for AI-Generated Inventions: The lack of clarity on whether AI-generated innovations qualify for patent protection creates investment risks. Without robust patent protections, companies may hesitate to invest in groundbreaking AI technologies, fearing unrestricted replication by competitors.

- Copyright Ambiguity for AI-Generated Works: Courts and regulatory bodies remain divided on whether AI-generated content qualifies for copyright protection. The absence of clear guidelines discourages businesses from leveraging generative AI for creative and software development applications.

- Trade Secret Vulnerabilities: Many AI companies rely on trade secrets to protect proprietary algorithms and training datasets. However, without stronger legal safeguards against model extraction attacks and unauthorized data access, trade secrets remain vulnerable to corporate espionage, cyber threats, and international IP theft.

- The Chilling Effect on AI Investment and Development: Uncertainty surrounding AI-related IP protections discourages venture capital funding and research initiatives. Investors prefer regulatory clarity before committing capital to new technologies, and without a strong legal framework securing AI IP rights, the U.S. risks ceding AI leadership to jurisdictions offering stronger and more predictable IP protections.

IV. Recommendations to Enhance IP, Privacy and Data Security Protection in AI Systems

A. Cybersecurity Enhancement Recommendations

To mitigate the risks set forth above, the AI Action Plan should address:

- Model-Specific Security Requirements. Establish security requirements based on model risk, including:
  - Security testing protocols for adversarial attack resistance.
  - Technical standards for detecting and preventing data poisoning attacks.
  - Cryptographic protections for model security.

- AI Supply Chain Risk Management Framework. Develop comprehensive supply chain security standards, including:

    - Vendor assessment protocols specific to AI components and services.
    - Model provenance verification requirements.

- Incident Response and Reporting Mechanisms. Create breach notification frameworks for AI systems that account for:
    - Differing incident types like model inversion attacks.
    - Reporting thresholds and timelines specific to AI deployment contexts.

B. Privacy Protection Recommendations

To mitigate the privacy risks set forth above, the AI Action Plan should address:

- Implementation of a Federal Privacy Law: the lack of a comprehensive federal consumer privacy law has created a fragmented regulatory environment where states—and even individual cities and municipalities—are introducing their own AI and data protection laws. This patchwork approach makes it increasingly difficult for data-driven AI businesses to operate efficiently in the U.S., as they must navigate an inconsistent and often contradictory web of privacy regulations across jurisdictions.

- Establishing Privacy by Design as a Cornerstone of AI Development: Privacy by design ensures that AI models and data processing systems are built with proactive privacy safeguards, rather than treating privacy as an afterthought. AI models should be developed with default privacy settings, ensuring that data minimization, encryption, and user control mechanisms are integrated from the start.

- Risk-Based Privacy Assessments: AI developers should conduct privacy impact assessments (PIAs) and risk evaluations throughout the AI lifecycle, documenting how data is collected, used, and protected.

- Enforceable Standards for Privacy-Preserving AI: Privacy by design and use of privacy enhancing technologies (PETs) should be encouraged. Recommended measures can include the following:

    - Specific technical parameters for differential privacy implementation that satisfy both innovation needs and legal requirements.
    - Data processing limitations aligned with purpose specification requirements.

- Operationalizing Data Minimization Across AI Lifecycles: AI developers should document data dependencies and justify data retention periods specific to model training, validation, and deployment phases.

- Transparency and Explainability Requirements. User trust in AI often depends on A decisions being transparent and explainable. Meaningful transparency requirements may include:

    - Consumer-facing disclosures explaining data usage, retention, and security measures to enhance public trust.

C. IP Protection Recommendations

To address the IP risks and concerns set forth above, the AI Action Plan should include:

- Patent and Copyright Protections for AI-Generated Works: Establish legal clarity on patentability and copyright ownership of AI-assisted innovations, ensuring that American businesses can secure rights to their intellectual property and protect their investments.

- Enforcement Against IP Theft and Misuse: Strengthen mechanisms for detecting and preventing unauthorized use, replication, or distillation of AI models and training data. This includes enhancing legal recourse against IP theft, increasing enforcement actions against foreign entities engaging in AI-related economic espionage, and collaborating with allies to establish international AI IP protections.

- Promoting Secure AI Knowledge Sharing: Develop frameworks for safe collaboration in AI research while maintaining robust safeguards against intellectual property theft and model extraction. Establish federal guidelines for AI licensing agreements, open-source best practices, and secure data-sharing protocols to enable responsible AI innovation without exposing American models to exploitation.

V. Leveraging Blockchain Technology for IP, Cybersecurity and Privacy Protection

In addition to the recommendations provided above, we also recommend that the U.S. explore ways to leverage blockchain technology in combination with AI to mitigate the intellectual property, privacy and cybersecurity risks often associated with AI systems. Blockchain's decentralized and immutable structure offers a transparent, and tamper-resistant framework for AI governance, mitigating risks related to data integrity, security breaches, algorithmic bias, and centralization concerns. As highlighted in my Forbes.com article on AI and blockchain integration, this synergy can potentially address AI's key vulnerabilities while fostering innovation and user trust.

President Trump's Executive Order on American Leadership in Digital Financial Technology emphasized the need for blockchain-driven security enhancements to bolster American financial strength and stability. Furthermore, the establishment of a U.S. Strategic Bitcoin Reserve reflects the recognition that decentralized technologies like blockchain are critical to maintaining America's economic and technological leadership.

Applying these principles to AI, the U.S. AI Action Plan should incorporate blockchain-based security frameworks to:

- Enhance AI Data Integrity and Security: Blockchain's immutable ledger helps ensure that AI model training data, updates, and decision-making processes are tamper-resistant and verifiable, mitigating risks such as data poisoning attacks and adversarial manipulations.

- Strengthen AI Supply Chain Resilience: The AI development lifecycle relies on multiple third-party datasets and cloud computing infrastructures, which introduce risks of unauthorized modifications and cyber threats. Blockchain-enabled AI model tracking can help make each stage of an AI system's lifecycle—from data collection to deployment—auditable and potentially more secure.

- Combat Deepfakes and AI-Generated Misinformation: Blockchain can serve as a trusted registry for authentic digital content, helping to distinguish AI-generated disinformation from verifiable sources. This is particularly critical as foreign adversaries seek to weaponize AI-generated deepfakes to influence American political discourse.

- Enable Privacy-Preserving AI with Zero-Knowledge Proofs (ZKPs): Privacy remains a significant concern in AI development. Blockchain-powered AI systems can leverage ZKPs to verify AI transactions and computations without exposing underlying sensitive data, helping to promote both compliance with privacy regulations and enhanced user trust.

- Automate AI Governance and Compliance through Smart Contracts: Blockchain-enabled smart contracts can help automate and streamline compliance with AI privacy guidelines, IP protections, and security standards.

- Track Provenance and Protect Intellectual Property: One of the key challenges of AI innovation is IP infringement and model replication by unauthorized actors. Blockchain's ability to establish an immutable record of AI-generated content, model training data, and proprietary algorithms allows for verifiable provenance tracking, ensuring that AI developers and businesses can protect their intellectual property rights.

VI. Conclusion

A strong federal AI Action Plan must prioritize privacy, cybersecurity, intellectual property protections, and regulatory flexibility to ensure U.S. leadership in AI. Establishing a governance framework that both protects user rights and fosters innovation will be key to maintaining America's strategic advantage in artificial intelligence.

I welcome the opportunity to contribute further insights. Thank you for your consideration.